Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

III SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
END SEMESTER EXAMINATIONS, NOVEMBER 2019

ELECTRICAL CIRCUIT ANALYSIS [ELE 2153]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 15 November 2019** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. For the circuit shown in Fig.Q1A, draw the Thevenin`s equivalent circuit across R_L. ***(04)***

1B. For the circuit shown in Fig.Q1B, find the current through (3+j4) Ω using Superposition theorem. ***(03)***

1C. Determine the output response y(t), when the impulse response h(t) for an input x(t) is as shown in Fig.Q1C. ***(03)***

2A. Determine whether the following signal is energy or power signal and also find the energy and the power of the signal.

$$x(t) = \begin{cases} 2\left(1 - \left|\frac{t}{4}\right|\right) & ; for\ |t| \le 4 \\ 0 & ; otherwise \end{cases}$$

(03)

2B. The first derivative of the function f(t) is shown in Fig.Q2B. Obtain the equation for f(t) and sketch it. Also, determine the value of 'K' so that $\int_{-\infty}^{\infty} f(t)\,dt = 0$. ***(03)***

2C. For the network shown in Fig.Q2C, switch is moved from position 1 to 2 at t = 0. Switch is in position 1 for a long time. Determine i(t) for t>0 using time domain analysis. ***(04)***

3A. Find the Laplace transform of the periodic waveform shown in Fig.Q3A. ***(03)***

3B. Draw the pole-zero diagram defined by the function $F(s) = \frac{8(s+2)(s^2+9)}{(s^2+4s+3)(s^2+2s+5)}$. Using pole-zero diagram, find the residues at all poles. Hence find f(t). ***(03)***

3C. For the circuit shown in Fig.Q.3C, find the voltage $v_o(t)$ using Laplace Transform technique. The circuit is under steady state for t<0 and the switch is opened at t=0. ***(04)***

4A. Determine the admittance parameters of the two-port network shown in Fig.Q4A. Hence, compute the values of the parameters at s = j0.5 rad/sec. ***(05)***

4B. Two networks A and B are connected in parallel. If $[Z_A] = \begin{bmatrix} 14 & 4 \\ 20 & 6 \end{bmatrix}$ and $[Y_B] = \begin{bmatrix} 1/s & -1/s \\ -1/s & 1/s \end{bmatrix}$, find the Transmission (ABCD) parameters of the overall network. ***(03)***

4C. Derive the h-parameters of the two port circuit shown in Fig.Q4C. ***(02)***

5A. Write the exponential Fourier Series of the waveform shown in Fig.Q5A. Also, plot the magnitude and phase spectra. ***(05)***

5B. Find the Fourier transform of the function $f(t) = t.\, e^{-2t}\, u(t-2)$. ***(03)***

5C. Find the energy of the signal shown in Fig.Q5C in frequency domain using Parseval`s Theorem. ***(02)***

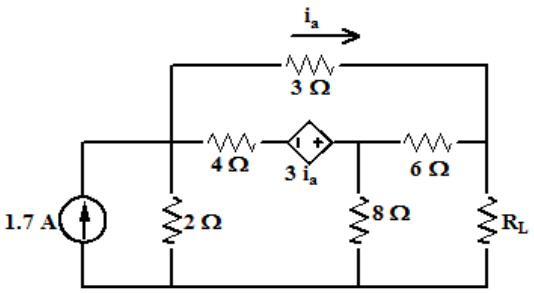


Fig.Q1A

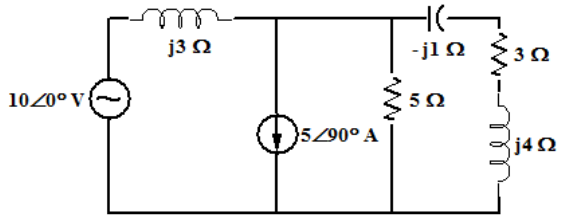


Fig.Q1B

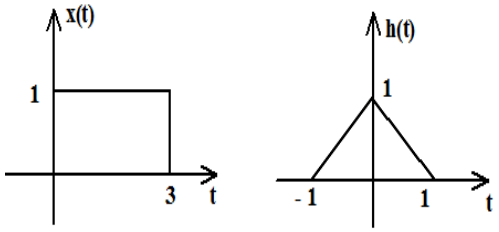


Fig.Q1C

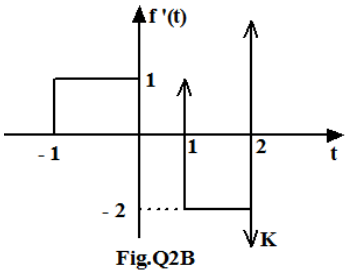


Fig.Q2B

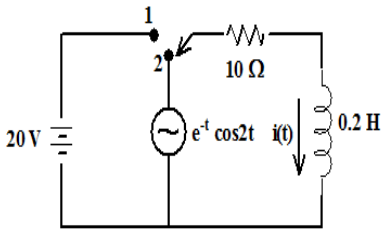


Fig.Q2C

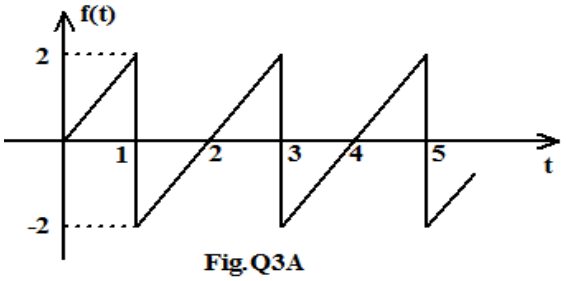


Fig.Q3A

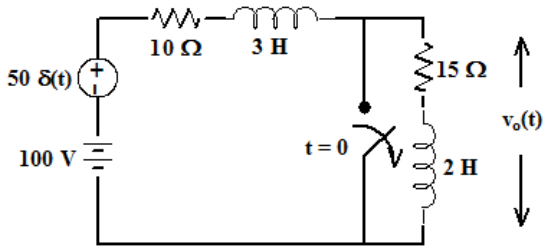


Fig.Q3C

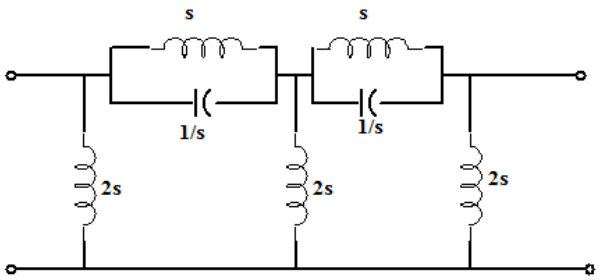


Fig.4A

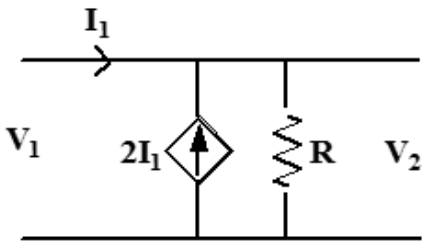


Fig.Q4C

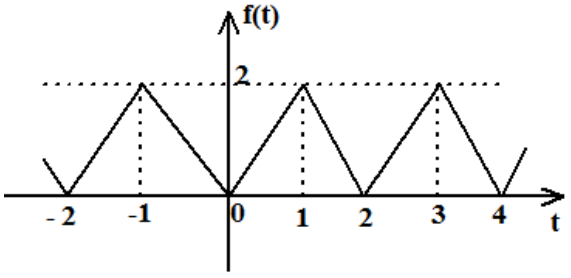


Fig.Q5A

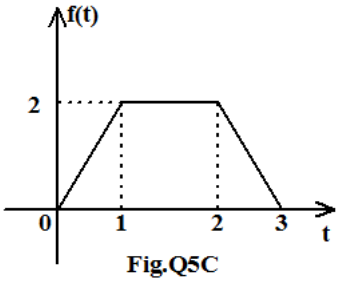


Fig.Q5C